

Exploring double categories of relations

Hayato Nasu

Kyoto Univ.

this Sep.



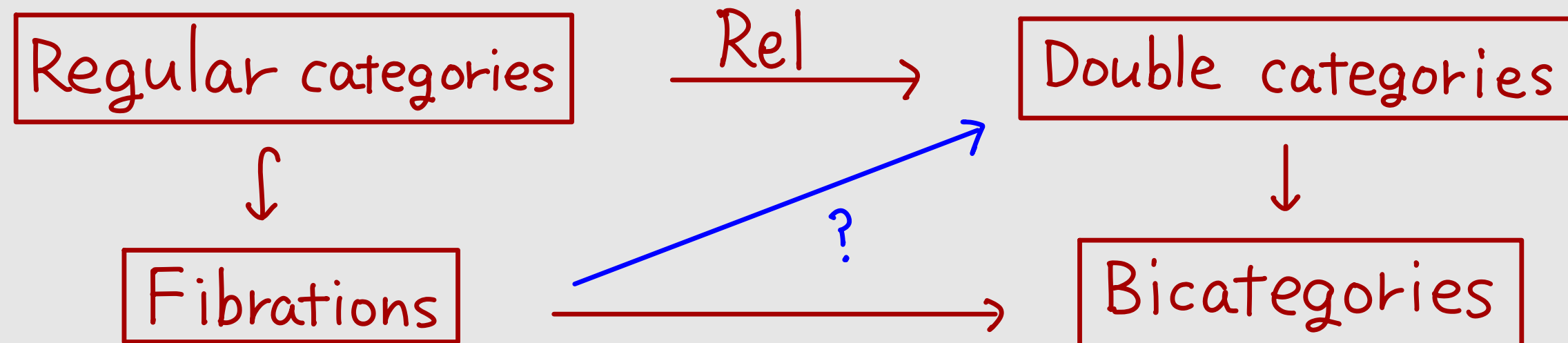
Dalhousie Univ.

FMCS 2025

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Category-theoretic understanding of relations:

- has been investigated in the **bicategorical** setup for a long time.
- has been studied in relation to **hyperdoctrines & fibrations**.
- has been approached using **double categories** more recently [Lambert'23].



The guiding question :

What makes a double category of relations?



Rel , $\text{Rel}(\mathcal{C})$, Span ,
 $\text{Rel}_{(E,M)}(\mathcal{C})$, $\mathcal{Q}\text{-Rel}$, ...



Prof , Par , ...

My answer : the Frobenius axiom

1. Background
2. The Frobenius axiom
3. Bonus : Maps revisited

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2. The Frobenius axiom

3. Bonus : Maps revisited

Bicategories of relations

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Allegory

[Freyd, Scedrov, '90]

Formulation in terms of *involution*.

$$(-)^{\circ} : B(I, J) \longrightarrow B(J, I)$$

Cartesian bicategory

[Carboni, Walters '87, Carboni, et.al. '07,
Todd, nLab post.]

Formulation using *adjoints*
in bicategories.

Theorem. [Freyd, Scedrov]

B is a unital
tabular allegory $\iff B \simeq \text{Rel}(\mathcal{C})$
for a regular category $\mathcal{C}$ $\iff B$ is a functionally-
complete discrete
locally-posetal
cartesian bicategory.

Bicategories of relations, generalized

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Definition. $p : \mathcal{E} \rightarrow \mathcal{B}$: fibration, $\mathcal{B}$ has finite products, $I, J \in \mathcal{B}$

| A p -relation $I \rightarrowtail J$ is an object $R \in \mathcal{E}_{I \times J}$. the fiber over $I \times J$

Proposition. [Pavlovic '96, Lawler '15]

| $p : \mathcal{E} \rightarrow \mathcal{B}$: a regular fibration $\rightsquigarrow R_p$: a bicategory of p -relations.

Existence of a (partial) right adjoint is stated without proof.

Theorem. [Bonchi, Santamaria, Seeber, Sobocinski '21]

| $\left(\begin{array}{c} \text{elementary existential} \\ \text{doctrines} \end{array} \right) \xrightleftharpoons[\perp]{R-} \left(\begin{array}{c} \text{'bicategories of} \\ \text{'relations' in [CW '87]} \end{array} \right)$

Double categories

Definition.

A (pseudo) double category consists of

- objects $A, B, \dots$

- vertical arrows $A \downarrow f B, \dots$

- horizontal arrows $A \xrightarrow{p} B, \dots$

- cells $\begin{array}{ccc} A & \xrightarrow{p} & B \\ f \downarrow & \sigma & \downarrow g \\ C & \xrightarrow{q} & D \end{array}, \dots$

with compositions of vertical arrows,

$$\begin{array}{c} A \\ \downarrow f \\ B \\ \downarrow g \\ C \end{array}$$

horizontal arrows,

$$A \xrightarrow{p} K \xrightarrow{m} X$$

and cells

$$\begin{array}{ccccc} A & \xrightarrow{p} & K & \xrightarrow{m} & X \\ f \downarrow & \sigma & \downarrow s & \nu & \downarrow u \\ B & \xrightarrow{q} & L & \xrightarrow{n} & Y \end{array}$$

$$\begin{array}{ccccc} A & \xrightarrow{p} & K & & \\ f \downarrow & \sigma & \downarrow s & & \\ B & \xrightarrow{\quad} & L & & \\ g \downarrow & \tau & \downarrow t & & \\ C & \xrightarrow{r} & M & & \end{array}$$

subject to some coherence conditions.

Double categories

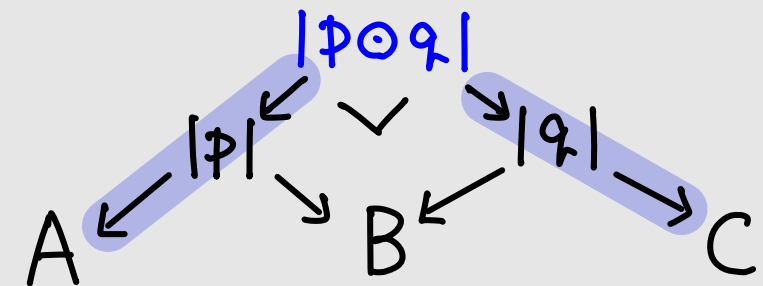
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<u>Example</u>	Rel	Span	Prof
Objects	sets	sets	categories
Vertical arrows	functions	functions	functors
Horizontal arrows	relations	spans	profunctors
Cells	$ \begin{array}{ccc} A & \xrightarrow{p} & B \\ f \downarrow & \tau & \downarrow g \\ C & \xrightarrow{q} & D \end{array} $ A cell exists iff $ \begin{array}{c} p(a, b) \\ \Downarrow \\ q(f(a), g(b)) \end{array} (\forall a, b) $	$ \begin{array}{ccc} A & \xleftarrow{ p } & B \\ f \downarrow & \eta & \downarrow h & \eta & \downarrow g \\ C & \xleftarrow{ q } & D \end{array} $	$ \begin{array}{ccc} A^{\circ p} \times B & \xrightarrow{p} & \text{Set} \\ f^{\circ p} \times g \downarrow & \Downarrow \tau & \\ C^{\circ p} \times D & \xrightarrow{q} & \end{array} $

Remark

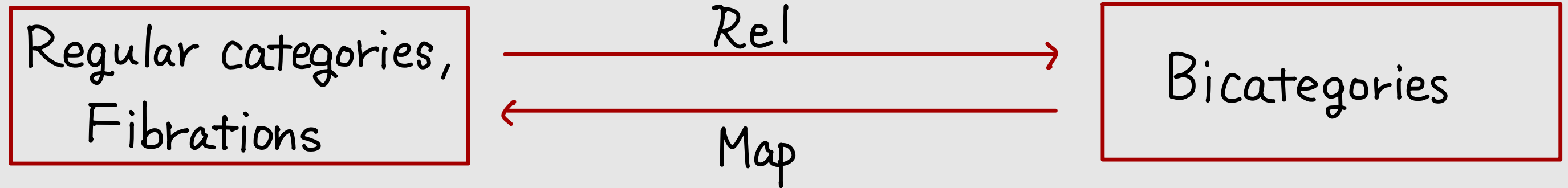
The horizontal composition is required to be associative *only up to isomorphism*.

e.g. the composition of spans



Motivating double categories

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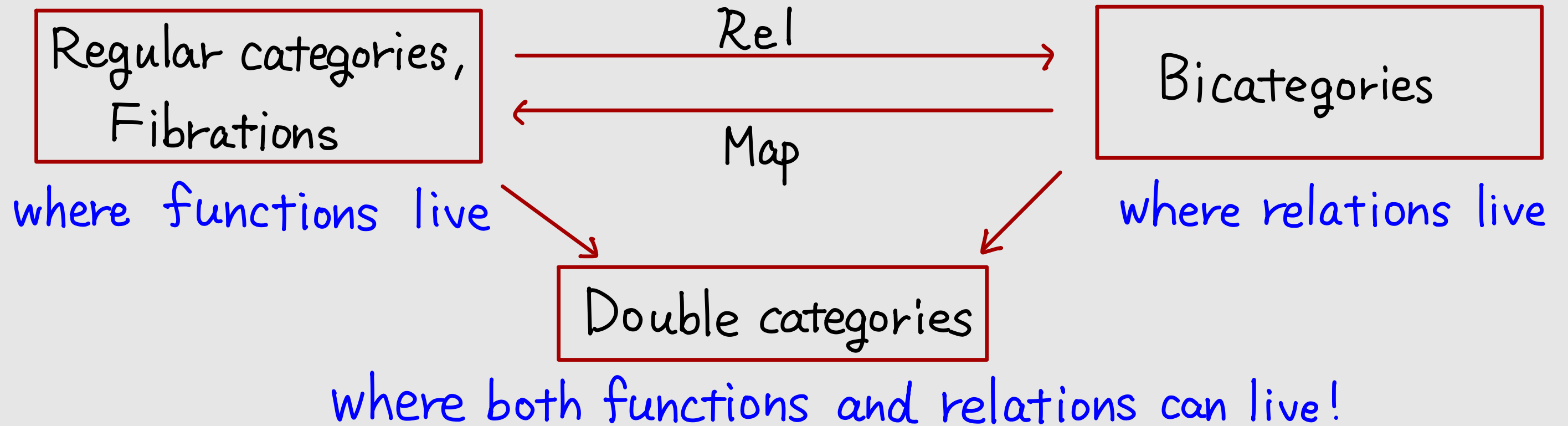
Motivating double categories

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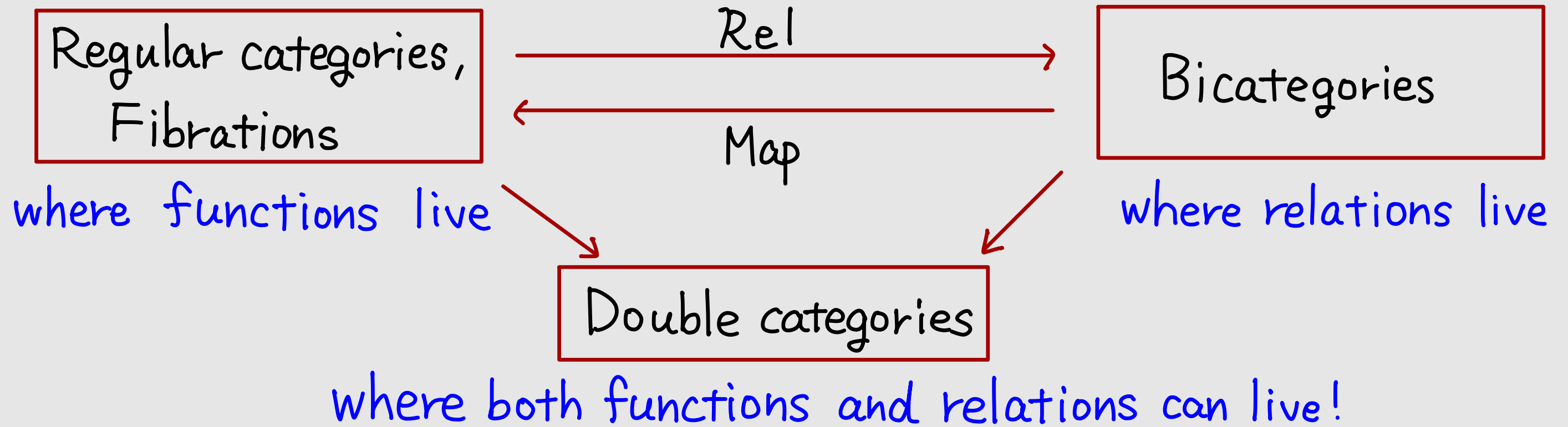
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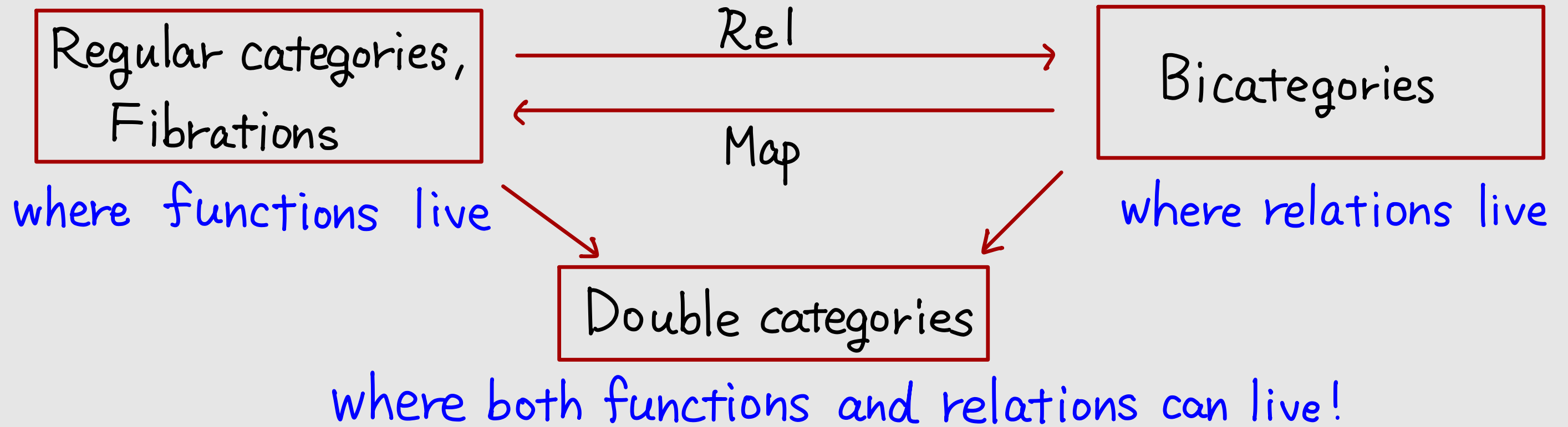
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- ① can express the interaction b/w functions & relations with companions & conjoints.

Motivating double categories

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- ① can express the interaction b/w functions & relations with companions & conjoints.
- ② can distinguish functions from maps ($\doteq$ functional relations)

Motivating double categories

③ Cartesian double categories are easier to handle.

Definition. $\mathbb{D}$: double category

$\mathbb{D}$ is cartesian if $\mathbb{D} \begin{array}{c} \xrightarrow{!} \\ \perp \\ \xleftarrow{\exists} \end{array} \mathbb{1} \quad \& \quad \mathbb{D} \begin{array}{c} \xrightarrow{\Delta} \\ \perp \\ \xleftarrow{\exists \times} \end{array} \mathbb{D} \times \mathbb{D} \quad \text{in } \mathbf{Dbl}.$

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Proposition. [Patterson '24, N.]

$\mathbb{D}$: cartesian equipment $\implies \mathcal{H}(\mathbb{D})$: a cartesian bicategory

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Remark Most of known examples of cartesian bicategories arise in this way. For a cartesian bicategory B , there is a cartesian double $\bullet\bullet$ bicategory whose horizontal part is B . [Verity'92]

1. Background

2. The Frobenius axiom

3. Bonus : Maps revisited

The Frobenius axiom

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Definition 2.1. (i) An object X in a Cartesian bicategory is discrete when the multiplication Δ_X^* and the comultiplication Δ_X satisfy

$$(D) \quad \Delta \cdot \Delta^* = (\Delta^* \otimes 1) \cdot (1 \otimes \Delta).$$

(We are forgetting the associativity in the middle.)

(ii) A Cartesian bicategory is called a 'bicategory of relations' if every object is discrete.

[Carboni, Walters. '87]

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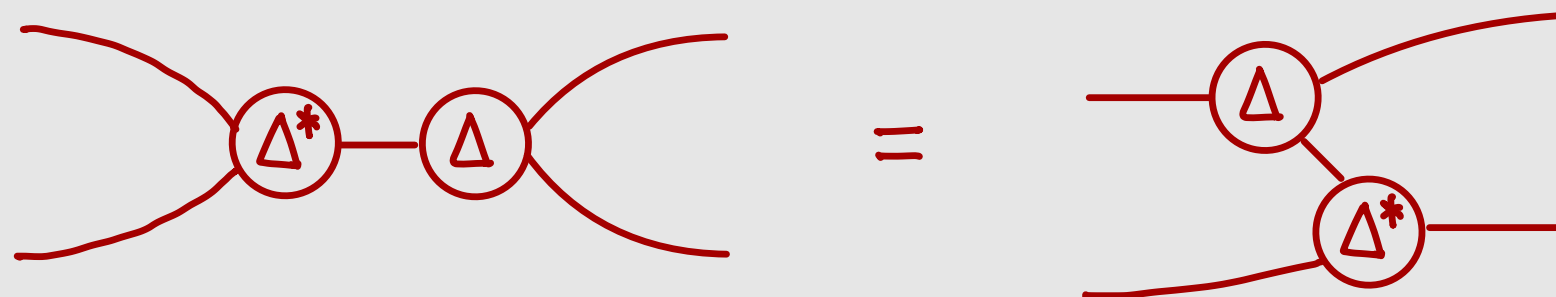
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This condition was later rephrased as the Frobenius axiom, because this is equivalent to the condition for Frobenius algebra [Walters, Wood '08]



The Frobenius axiom

Definition. [Walters, Wood '08, Lambert '23, Hoshino, N.'25, N.]

$\mathbb{D}$: cartesian equipment

$I \in \mathbb{D}$ is Frobenius if

$$\begin{array}{c}
 \begin{array}{ccc}
 & I & \\
 \Delta \swarrow & & \searrow \Delta \\
 I \times I & \xrightarrow{\text{opc.}} & I \times I \\
 \Delta \times \text{id} \searrow & \text{cart.} & \swarrow \text{id} \times \Delta \\
 & I \times I \times I &
 \end{array}
 =
 \begin{array}{c}
 \bullet \\
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 \end{array}
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 \Leftrightarrow \left(\begin{array}{ccccc}
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 \parallel & & \text{SII} & & \parallel \\
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canonically.

$\mathbb{D}$ is Frobenius if every object in $\mathbb{D}$ is Frobenius.

Example. $\mathcal{C}$: a regular category $\Rightarrow \text{Rel}(\mathcal{C})$ is Frobenius

For $A \in \mathcal{C}$ and $(x, y) : A \times A$, $(z, w) : A \times A$,

$$\exists u : A \ ((x, y) = (u, u) \wedge (u, u) = (z, w)) \iff (x, x, y) = (z, w, w)$$

Example. In Prof, $\mathcal{C}$ is Frobenius $\Leftrightarrow \mathcal{C}$ is a groupoid.

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Proof sketch [Walters, Wood '08]

$I \in \mathbb{D}$ is Frobenius if $\Delta^* \Delta_* \cong (\Delta \times \text{id})_* (\text{id} \times \Delta)^*$

$$(\Delta^* \Delta_*)((c_1, c_2), (d_1, d_2))$$

$\downarrow$

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$$\begin{aligned} (\Delta^* \Delta_*)((c_1, c_2), (d_1, d_2)) &= \int^{e \in \mathcal{C}} \mathcal{C}(c_1, e) \times \mathcal{C}(c_2, e) \times \mathcal{C}(e, d_1) \times \mathcal{C}(e, d_2) \\ &\downarrow \\ ((\Delta \times \text{id})_* (\text{id} \times \Delta)^*)((c_1, c_2), (d_1, d_2)) &= \mathcal{C}(c_1, d_1) \times \mathcal{C}(c_1, d_2) \times \mathcal{C}(c_2, d_2) \end{aligned}$$

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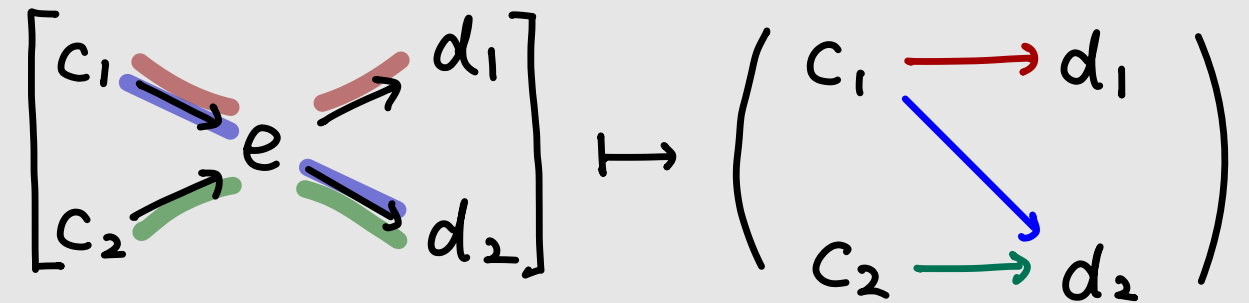
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$(\Rightarrow)$ Take $f: a \rightarrow b$ in $\mathcal{C}$.

$$\begin{array}{ccc} a & = & a \\ & \searrow f & \\ b & = & b \end{array}$$

$$\left[\begin{array}{ccc} c_1 & \xrightarrow{\text{red}} & d_1 \\ & \searrow \text{blue} & \\ c_2 & \xrightarrow{\text{green}} & d_2 \end{array} \right] \mapsto \left(\begin{array}{ccc} c_1 & \xrightarrow{\text{red}} & d_1 \\ & \searrow \text{blue} & \\ c_2 & \xrightarrow{\text{green}} & d_2 \end{array} \right)$$

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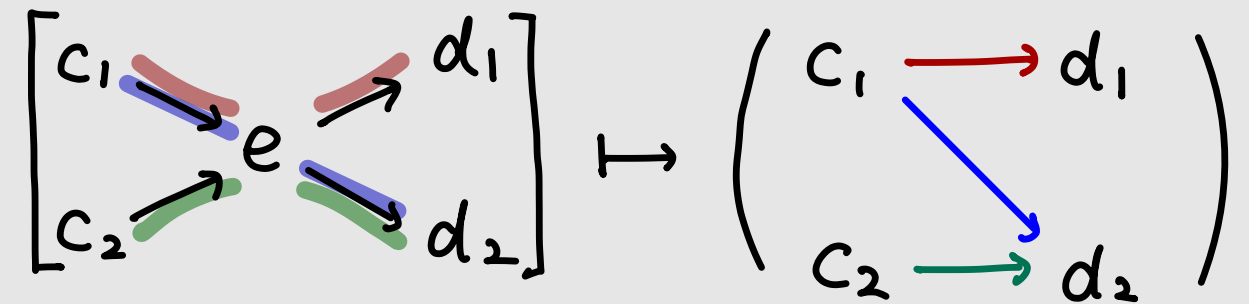
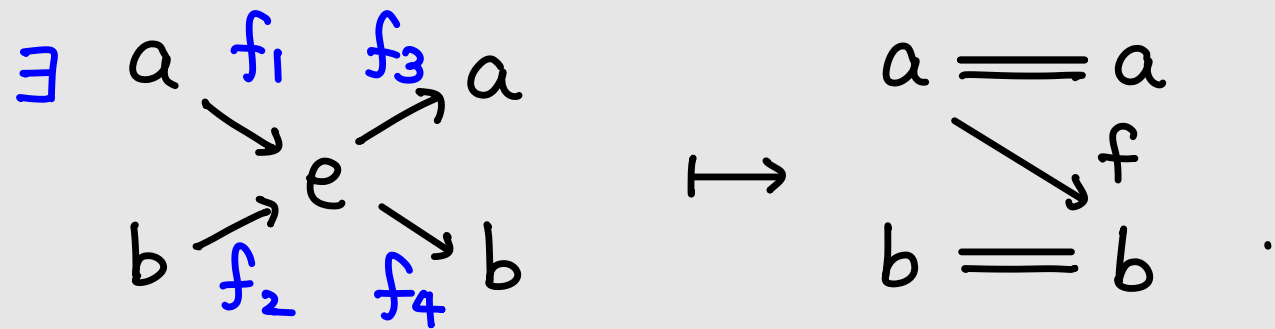
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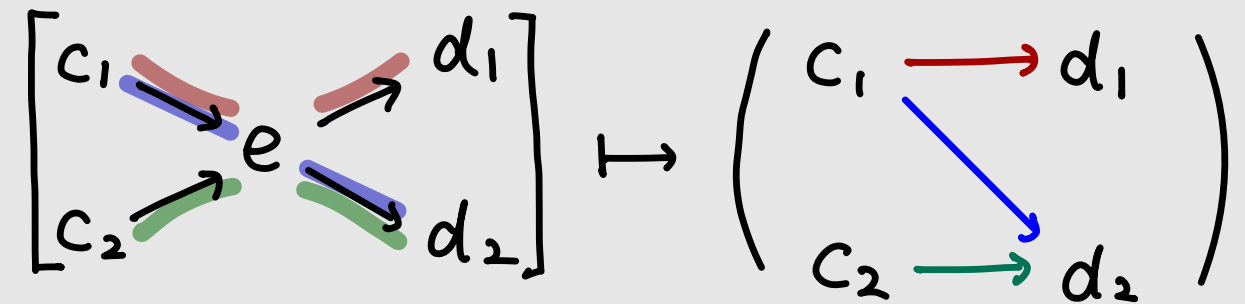
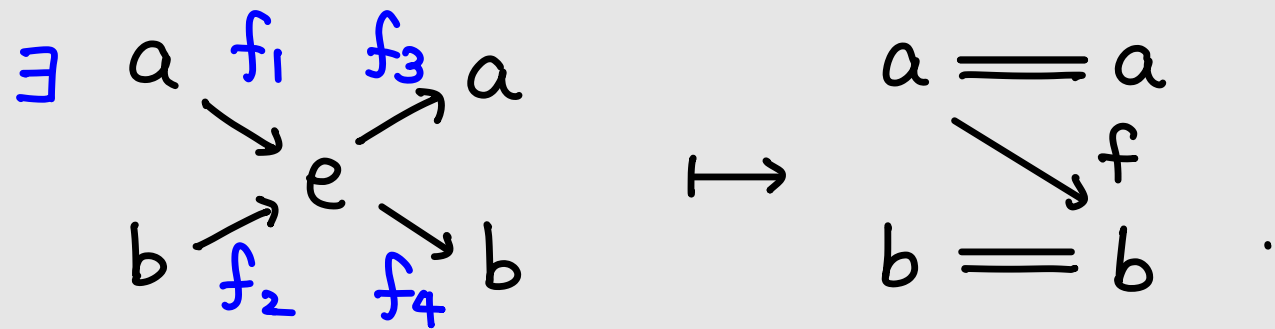
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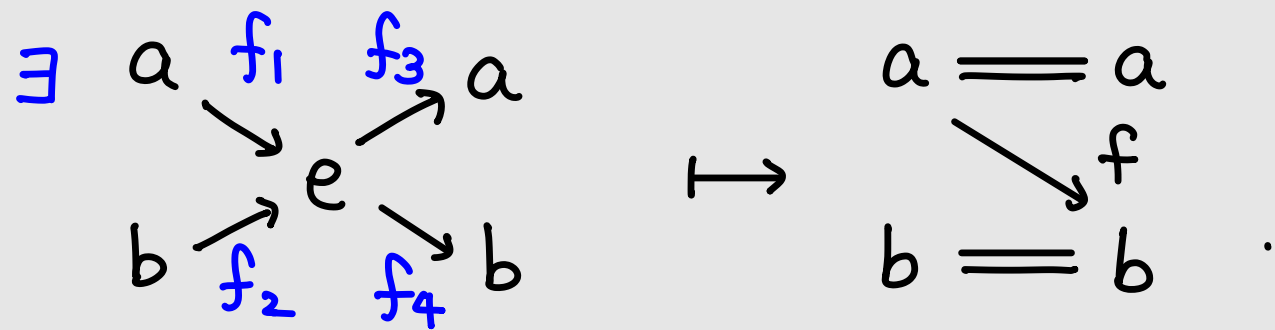
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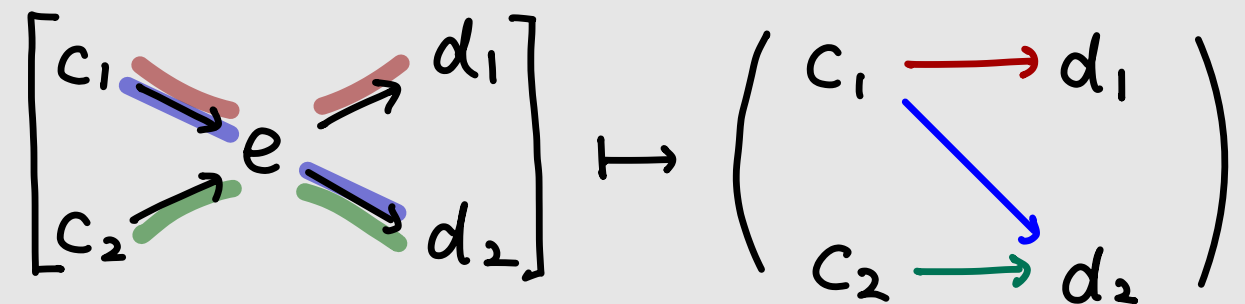
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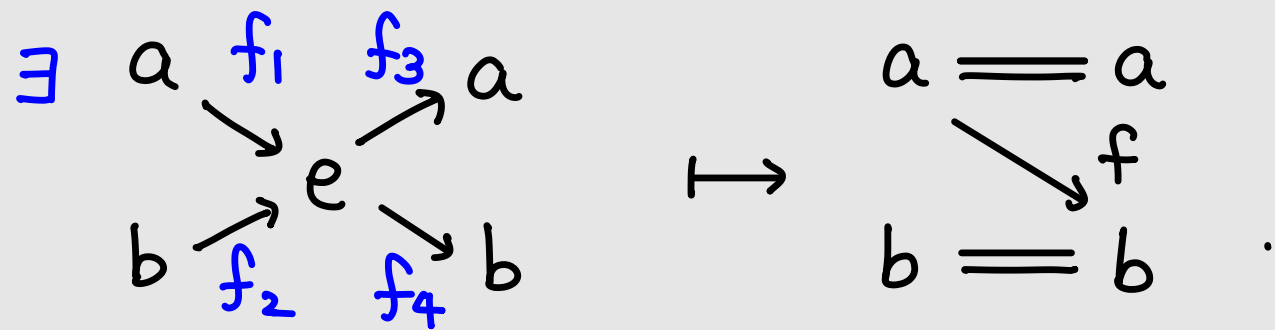
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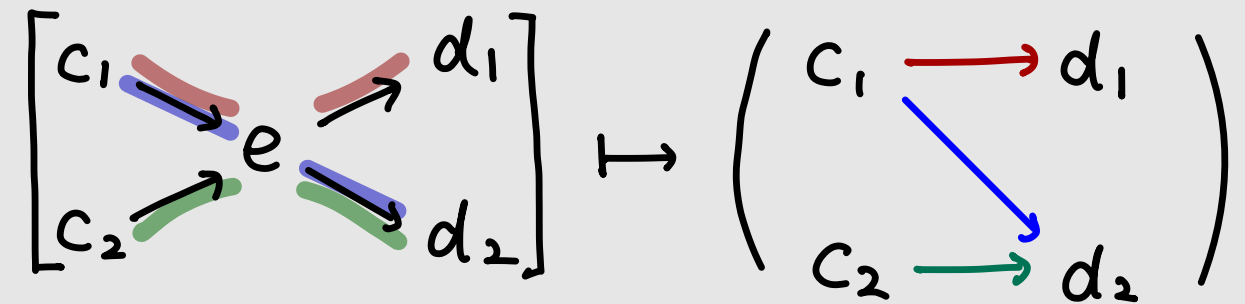
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The Frobenius axiom is an axiom of symmetry.

The Frobenius axiom induces involution

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Proposition. [Hoshino, N. '25, N.]

$\mathbb{D}$: a Frobenius cartesian equipment.

Every A and $\downarrow f$ in $\mathbb{D}$ are **self-dual** in the horizontal direction:

$$\mathbb{D}_h(C \times A, B) \cong \mathbb{D}_h(C, A \times B), \quad \mathbb{D}_c(h \times f, g) \cong \mathbb{D}_c(h, f \times g)$$

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The unit and counit on A are

$$1 \xrightarrow{!^*} A \xrightarrow{\Delta^*} A \times A \quad \text{and} \quad A \times A \xrightarrow{\Delta^*} A \xrightarrow{!^*} 1.$$

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$$1 \xrightarrow{!^*} A \xrightarrow{\Delta^*} A \times A \quad \text{and} \quad A \times A \xrightarrow{\Delta^*} A \xrightarrow{!^*} 1.$$

Cor. $\mathcal{H}(\mathbb{D})$ is a compact closed bicategory in the sense of [Stay '16].

From equipments to fibrations

Definition. $\mathbb{D}$: cartesian equipment $\rightsquigarrow$

a category $\mathcal{U}_{\mathbb{D}}$: objects : $A \xrightarrow{\alpha} 1$,

morphisms $(A \xrightarrow{\alpha} 1) \rightarrow (B \xrightarrow{\beta} 1) : \begin{array}{ccc} A & \xrightarrow{\alpha} & 1 \\ f \downarrow & \varphi & \parallel \\ B & \xrightarrow[\beta]{} & 1 \end{array}$

a fibration $\mathcal{U}_{\mathbb{D}} : \mathcal{U}_{\mathbb{D}} \longrightarrow \mathbb{D}_0 ; (A \xrightarrow{\alpha} 1) \mapsto A$

When $\mathbb{D}$ is Frobenius, all information in $\mathbb{D}$ is recoverable from $\mathcal{U}_{\mathbb{D}}$.

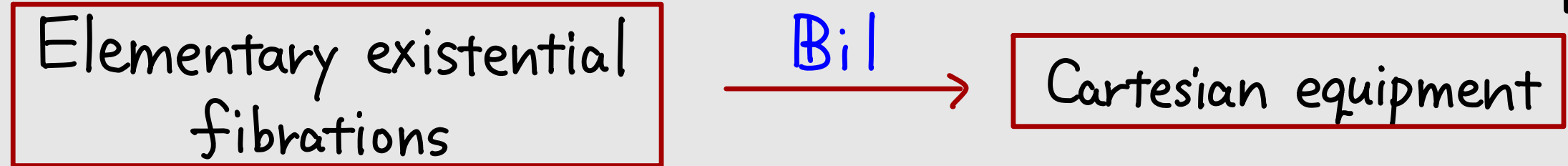
$$\begin{array}{ccc} \begin{array}{ccc} C & \xrightarrow{\alpha} & A \\ f \downarrow & \tau & \downarrow g \\ D & \xrightarrow[\beta]{} & B \end{array} & \rightsquigarrow & \begin{array}{ccc} C \times A & \xrightarrow{\tilde{\alpha}} & 1 \\ f \times g \downarrow & \tilde{\tau} & \parallel \\ D \times B & \xrightarrow[\tilde{\beta}]{} & 1 \end{array} & \rightsquigarrow & \tilde{\alpha} \xrightarrow{\tilde{\tau}} \tilde{\beta} \text{ in } \mathcal{U}_{\mathbb{D}} \end{array}$$

Proposition. [N.] $\mathbb{D}$: Frobenius $\Rightarrow$

$\mathcal{U}_{\mathbb{D}} : \mathcal{U}_{\mathbb{D}} \longrightarrow \mathbb{D}_0$ is an elementary existential fibration.

Main results

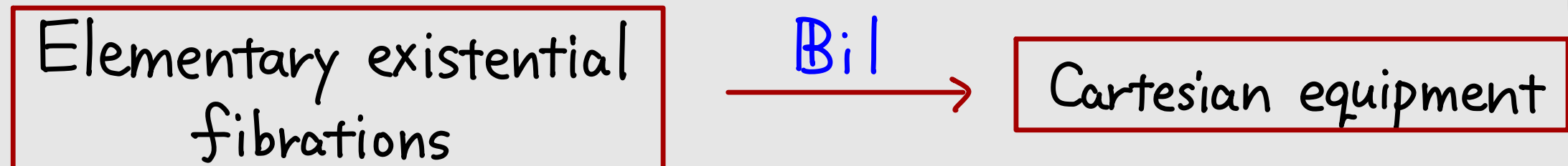
Pavlovic's bicategories of predicates are extended to double categories.
[Lawler '15, N.]



$\text{Bil}(\mathbb{P})$: the double category of $\mathbb{P}$ -relations

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$\text{Bil}(\mathcal{P})$: the double category of $\mathcal{P}$ -relations

Theorem. [N.]

$\mathcal{P}$: an e.e. fibration $\rightsquigarrow \text{Bil}(\mathcal{P})$: a Frobenius cartesian equipment.

This extends to a biequivalence:

$$\mathcal{E}\mathcal{E}\text{Fib} \xrightleftharpoons[\cong]{\text{Bil}} \text{CartEq}_{\text{Frob}}$$

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Example. $\text{Prof}_{\text{grpd}} \simeq \text{Bil}(\text{GrpdAct} \rightarrow \text{Grpd})$

1. Background

2. The Frobenius axiom

3. Bonus : Maps revisited

Revisiting maps

15 / 17

Definition. A **map** in a bicategory is a left adjoint 1-cell in it.

Revisiting maps

15 / 17

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Regular categories

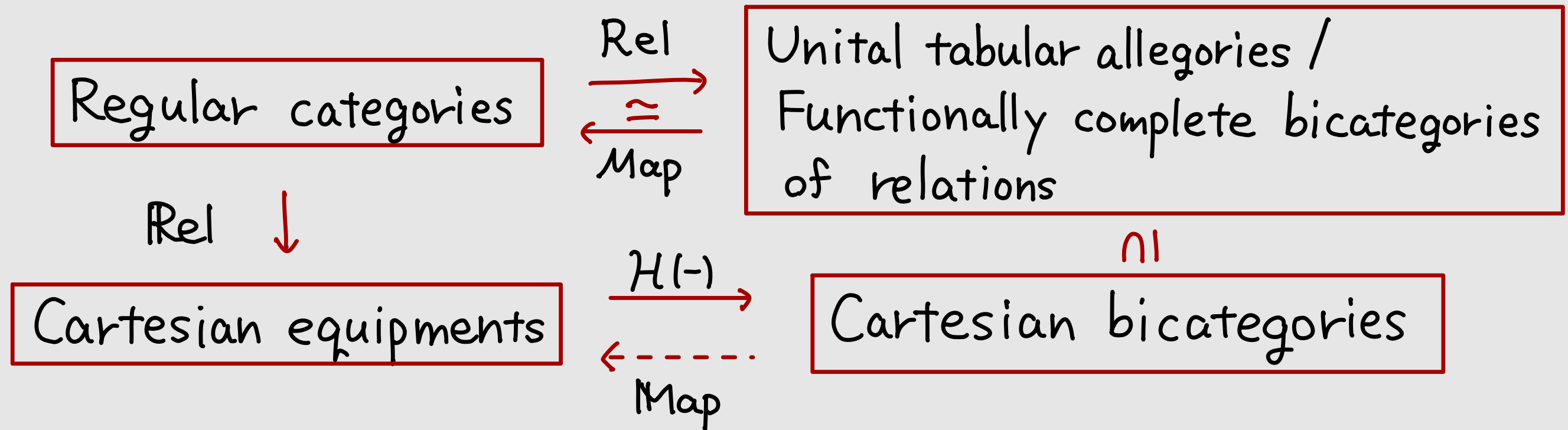
$\begin{array}{c} \text{Rel} \\ \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \text{Map} \end{array}$
 $\simeq$

Unital tabular allegories /
Functionally complete bicategories
of relations

Revisiting maps

15 / 17

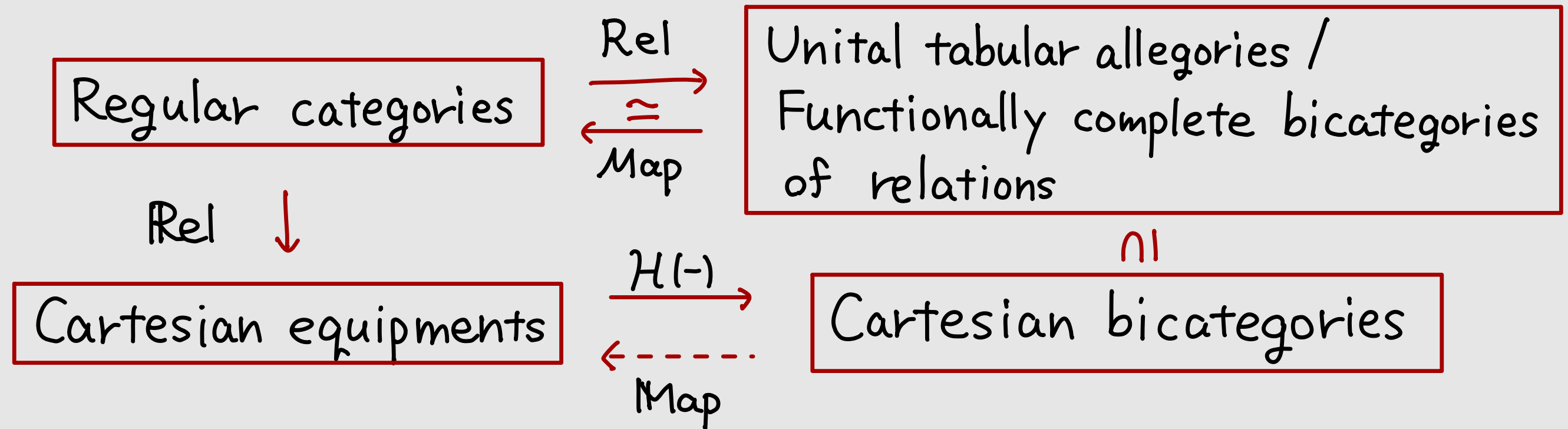
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Revisiting maps

15 / 17

Definition. A **map** in a bicategory is a left adjoint 1-cell in it.



Definition. A bicategory B is **map discrete** if $\text{Map}(B)(I, J) \subseteq \mathcal{B}(I, J)$
| is equivalent to discrete categories.

This is sufficient to construct a double category $\mathbb{M}\text{ap}(B)$.

Revisiting Map

16 / 17

Definition. [Paré 21] An equipment $\mathbb{D}$ is **Cauchy** if every map
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Revisiting Map

16 / 17

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Map discrete
Cauchy cart. equipments

$\xrightarrow{\mathcal{H}(-)}$
 $\xleftarrow{\text{Map}(-)}$
 $\cong$

Map discrete
cart. bicategories

Revisiting Map

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Map discrete
 Cauchy cart. equipments

$$\begin{array}{c} \mathcal{H}(-) \\ \xrightarrow{\quad} \\ \text{Map}(-) \end{array}$$

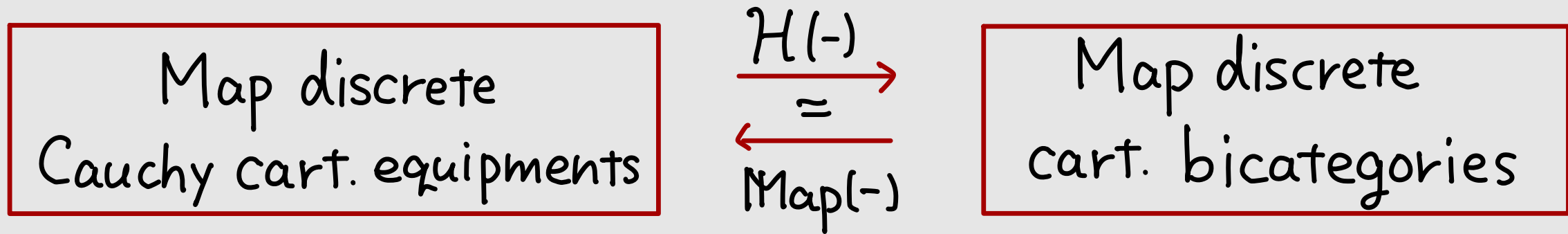
Map discrete
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Proposition. [N.]

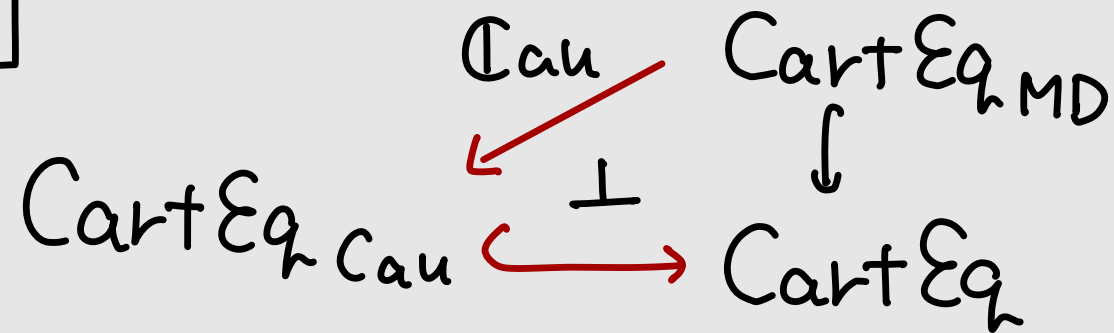
$$\begin{array}{ccc} & \text{Cau} & \text{CartEq}_{MD} \\ & \swarrow & \downarrow \\ \text{CartEq}_{Cau} & \xrightarrow{\perp} & \text{CartEq} \end{array}$$

Revisiting Map

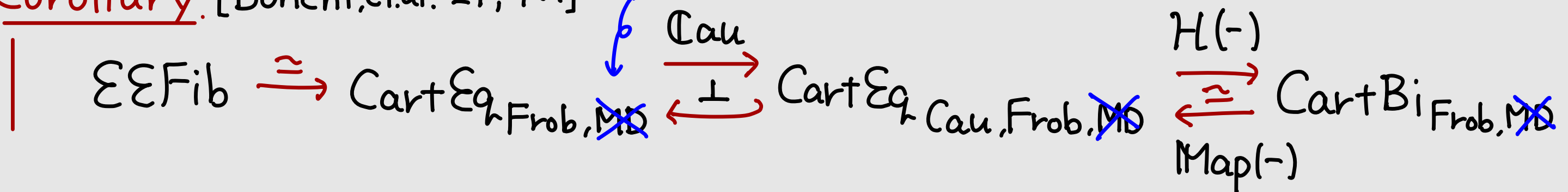
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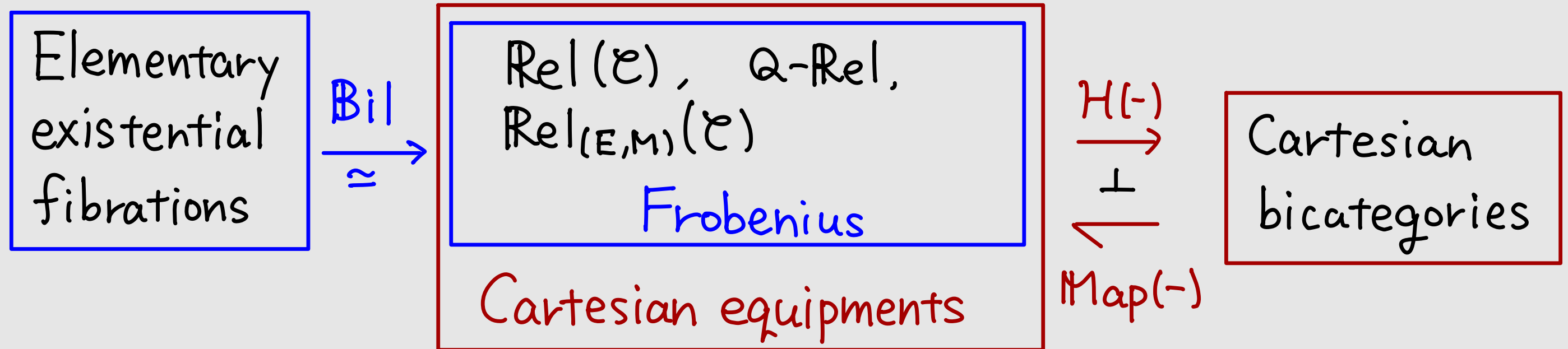
Proposition. [N.]



Corollary. [Bonchi, et al. '21, N.] [Pavlovic '96]



Summary



Future Work

- Generalizing to monoidal equipments.
- Studying categorical logic with cartesian equipments (quotient completion, tabulators)

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Thank you!

H.Nasu . Logical Aspects of Virtual double categories , master's thesis, 2025

arXiv : 2501.17869

Bicategories of relations

A1 / A2

- 1984 • "Bicategories of spans and relations" Carboni, Kasangian, Street.
defined $\text{Span}(\mathcal{C})$ & $\text{Rel}(\mathcal{C})$
- 1987 • "Cartesian bicategories I" Carboni, Walters.
- 1990 • "Categories, Allegories" Freyd, Scedrov.
- 1996 • "Maps II: Chasing diagrams in categorical proof theory" Pavlovic
defined bicategories of predicates for regular fibrations
- 2021 • "On Doctrines and Cartesian Bicategories" Bonchi et.al.

Double categories of relations and spans

A2 / A2

- 1999 + "Limits in double categories". Grandis, Paré.
- 2012 + "Span, cospan, and other double categories" Niefield.
- 2019 + "Cartesian Double categories with an Emphasis on Characterizing Spans" Aleiferi. characterized $\text{Span}(\mathcal{C})$
- 2023 + "Double categories of relations" Lambert
characterized $\text{Rel}(\mathcal{C})$ for regular categories $\mathcal{C}$
- 2025 + "Double categories of relations relative to factorization systems" Hoshino, N. characterized $\text{Rel}_{(E,M)}(\mathcal{C})$.